AI in Practice: Final Project

Design and build your own AI practice case

For the final project, pick a problem you actually care about and turn it into a small working AI system. It does not have to be novel research. It has to be yours, it has to run, and it has to show something you can measure.

1. Groups

Form a group of **2–3 students**, and agree on who coordinates the work.

2. Topic

You can either extend a direction from class or propose your own idea. There is no fixed list of acceptable topics.

- **From class:** object detection and edge deployment; vision-guided robotic grasping; LLMs, VLA and agents; intelligent systems and smart healthcare; humanoid robots and motion control; AI safety and security.
- **Your own idea:** any application where AI does real work — campus, lab, sports, music, health, accessibility, and so on.

Aim for something you can get running in a week or two. A small project that works is worth more than a large one that almost works. If you are unsure whether a topic is too big or too small, describe it in a few lines and ask.

3. What counts as innovation

Keep this modest. Any one of the following is enough:

- a new scenario or your own dataset;
- a combination of two methods covered in class;
- a better interaction or a clearer interface;
- a careful evaluation of something that is usually only demonstrated.

Using existing models, open-source code and APIs is completely fine. What matters is that you can say what you changed and compare it with a simple baseline.

4. What to present

Slides. There is no fixed length — use as many as the talk needs. They should cover: topic and motivation; team and roles; problem and data; approach; results and demo; limitations and what you learned.

A live demo or a short screen recording. One to two minutes is plenty. A failed case is welcome — it usually shows more understanding than a perfect one.

No written report and no code submission. You do not need to hand anything in. The project is assessed through the presentation, so make sure the work is visible in the slides and the demo.

5. Presentation

15 minutes for the talk, including the demo, followed by **3–5 minutes** of questions. **The whole slot must not exceed 20 minutes.** How you divide the talking is up to the group.

6. Grading

What is assessed	Weight	What we look for
Problem and idea	25	A clear, relevant problem with a sensible scope
Implementation	35	The system runs; results are compared with a baseline
Presentation	25	Clear slides, understandable in 15 minutes, questions answered
Teamwork	15	Clear roles; the work is shared across the group

There is no penalty for a simple method, a small dataset or results that fall short of expectations, as long as the work is honest and the reasons are discussed.

7. A few rules

- Cite the data, models and code you use, and state how you used AI assistants.
- Report what you actually obtained. Fabricated or edited results are treated as misconduct.
- Work on attacks or privacy should stay inside your own or authorised environments.

8. Schedule

Because of the National Day holiday, one lecture was cancelled: **AI safety and security is lecture 7**. The schedule for the end of the semester is therefore:

- **Lecture 7 (AI safety and security):** no additional practice task for this lecture. The class time is left to you, to form groups, settle on a topic and get the first step running.
- **From lecture 8:** presentations and questions, group by group.

If something does not work out, tell us early. Changing the topic or narrowing the scope is always allowed, and it is much easier before the last week than after.

We have a WeChat group for the course. Ask anything there at any time — that is the fastest way to reach us.

Instructors: Dr. Lili Fan, Dr. Yingwu Li

